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ABSTRACT

Microphotography for archives advocates the use of microfilm as a means of publication. The manual is an introduction for the archivist and provides definitions of microforms, microfilm, microfiche, microcards, and microprint. The uses and the advantages or disadvantages of both microforms and microfilm are presented. The manual also reviews archival operations, microfilm equipment, and storage and maintenance procedures of original negatives. (MM)

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MICROPHOTOGRAPHY FOR ARCHIVES

by

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PREFACE

This brief manual has been developed by the Microfilming Committee of the International Council on Archives to advance the use of microfilm as a means of publication with the hope that many archival institutions that are interested in facilitating access to their records for scientific and scholarly researchers as well as in preserving the records themselves will find it of value.

This manual is an introduction for the archivist rather than an exhaustive treatise. It has not been designed to be a guide for the technician although even he should find it useful. The cameras, readers, and other equipment illustrated are intended primarily to acquaint archivists with types of equipment. Items shown are not necessarily recommended.

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MICROPHOTOGRAPHY FOR ARCHIVES

1. Introduction

Microforms

The term microform covers all forms of micro-images -- these may be transparent or opaque, and they may be in the form of film rolls of various widths and lengths, in the form of film strips, or in the form of sheets or cards of various sizes.

All microforms are reduced photographic images of documents that must be magnified on a reader or viewer in order to be read.

The microforms in use today are basically four in number: (1) microfilm, (2) microfiche, (3) microcard, and (4) microprint.

Microfilm

Microfilm may be defined either as (1) a transparent flexible material for the photographic reproduction of documents in reduced size, or as (2) a series of photographic reproductions on this medium that may be v[illegible] optically and that may be used for making additional microfilm copies or enlarged to make eye-legible prints.

The microfilm in use today is of three types:

(1) the silver-halide emulsion type
(2) the diazo or ozalid type
(3) the Kalvar type

Emulsion film consists of a thin strip or roll of cellulose acetate, known as the base, containing a coating of light-sensitive emulsion on one side. Usually the emulsion side of microfilm has little or no gloss while the non-emulsion side is very glossy. The only permanent record films, i.e., those that have archival permanence, are the silver-halide emulsion type.

If we distinguish emulsion-type film according to the use made of the film there are four classes in all.

The first of these is the master, or camera or original negative. In this first generation copy the normal tonal values of the document micro-copied are reversed, that is black writing on white paper appears on the film as white on black. The master negative if it is to be preserved as an archival copy should be used only for making additional film copies. The only time it should be used in a microfilm reader or viewer is during the inspection process.

The master positive or security copy. In this second generation copy black ink on white paper appears on the film as a black image on a white background, that is, the microfilm copy has the same tonal values as the original text of the document that was photographed. This copy is specially prepared for use in producing a duplicate negative in case the master negative is damaged, lost, or destroyed.

The positive or reference copy. This copy, usually a second generation copy, also has the same tonal values as the original text photographed, i. e., black writing on white paper appears on the film as black on white. In the past few years, a film base made of polyester (mylar or cronar are two names for polyester), has been used by some manufacturers for making silver-halide microfilm in addition to the silver-halide positive film they manufacture on a cellulose acetate base. The polyester base film is thinner than acetate base film and is much stronger and more resistant to tearing.

The duplicate negative copy. This copy, usually a third generation copy, has the same tonal values as the master or camera negative, i. e., the normal values are reversed.

Just two years ago a new direct negative microfilm was produced by Eastman Kodak. This film makes it possible to produce a silver-halide duplicate negative from an original negative directly without going through the intermediate stage of preparing a positive print. This development makes it feasible to preserve the original negative as a "master" security copy and use the direct negative for production of positive prints.

Archivists should use only those films that are so composed and treated that the silver-halide image as well as the base meet technical standards for permanent record film. *

Ozalid or diazo type film consists of a thin strip of cellulose acetate that has a light-sensitive diazo dye either incorporated into the film base or coated on the base. When this film is exposed to ultraviolet and blue light and developed by an ammonia vapor or an alkaline solution an image is formed. With this process a negative image will produce a negative image.

Kalvar film consists of a thin strip of polyester (mylar or cronar) film that contains a diazonium emulsion. This film when exposed to ultraviolet

*Some Countries have standards for permanent record film. In the United States, for example, the U. S. Standard is identified as PHl. 28-1957. The International Standards Organization has not, as yet, recommended a standard for permanent record film.

light and then processed by heat, forms an image. With Kalvar film a negative image will also produce a negative image.

Kalvar and ozalid films are, at present, not suitable for use in a microfilm camera but only for preparing film duplicates. Neither one has been approved for archival permanence. Both ozalid and Kalvar film are generally more suitable for the reproduction of film copies from high contrast materials such as newspapers or line drawings than they are for the reproduction of archival materials.

The emulsion type microfilms, the ones suited for the reproduction of archival materials, have been especially developed. Some of the characteristics of archival grade microfilm negatives are fine or very fine grade emulsions with high resolving power, good sensitivity, enabling the reproduction in black, white, or greys, of the complete visible spectrum, and good contrast and latitude. Different emulsions are used depending upon whether the microfilm is negative or positive and whether the film is to be used in a rotary or a flat-bed camera.

Microfilm may be perforated (i. e., equipped with sprocket holes) along one edge or along both edges. Most microfilm cameras in use in the United States today use unperforated film. Some European cameras still use perforated film although the trend is toward the use of non-perforated film as the use of perforated film results in the loss of 25 percent or more of the useful area of the film. This is so because the image on the film must be kept in the area between the perforations.

Most microfilm produced today of archival materials, newspapers, and books is 35mm. on reels with a capacity of not more than 110 feet (33.53 meters). The standard roll length is 100 feet (30.48 meters) but rolls are often made shorter, or a little longer, to meet bibliographical or archival considerations. For example, it is not desirable when filming a series of bound volumes or dossiers to have part of one volume or dossier on one roll and another part of the volume or dossier on another roll. Therefore, breaks are normally made at the end of a volume or a dossier. The length of the resulting film roll will reflect this. 16mm. film is occasionally used by archives for the reproduction of card indexes and some letter size materials with good contrast between the writing and the paper base. Occasionally 70mm. or even 105mm. microfilm is used for the reproduction of large engineering or architectural drawings that cannot be microfilmed satisfactorily on 35mm. film.

During early stages of the development of microfilming when the filming of very short runs of material was quite common, microfilm was, in some cases, kept in the form of strips, especially in France. The maintenance and use of microfilm in strip form is declining sharply because of difficulties in storing and handling.

Microfilm may, however, be cut into strips and inserted in acetate jackets (Figure 1). Eye-legible identifications can be typed on the top of legibility. The modern step and repeat cameras, used for microfiche reproduction, cost many times more than a modern planetary camera, and are the only ones whose speed and efficiency approaches that of a modern planetary microfilm camera. The reduction ratios generally used for microfiche (i. e. 18:1 or 20:1) are not only too high for the satisfactory reproduction of some archival materials but the placement of images of archival materials within a grid framework is often impracticable.

Filming errors in microfiche production usually require redoing of the entire fiche; on roll film retakes may be spliced onto a roll in their proper position much more easily. Retakes are usually more prevalent in archival filming than in the reproduction of those materials suitable for microfiche or microcard reproduction.

Although good microfilm readers are usually a little more expensive than microfiche or microcard readers this cost factor is more than compensated for by the fact that a good microfilm reader generally has a larger screen to accommodate larger images as well as a revolving head that will permit easy viewing of images in all four microfilm placements. In contrast, most screens of microcard or microfiche readers are not only small but also images on microfiche or microcards that are not in right reading positions may be adequately viewed only if the fiche or card is removed from the reader, turned at right angles and reinserted. When using fiche or cards that have numerous images of this kind it is often most difficult for the viewer to avoid confusion.

Microfilm is much more suitable for copying extensive series of documents. A 100 foot (30.48 meters) roll of microfilm can contain many times the number of documents that can be contained on a fiche or on a card. The microfilm format, therefore, is more adaptable to the handling of larger units. Consequently less work is required to title and describe its contents.

There are, of course, some materials in archives that lend themselves to reproduction on microfiche or mocrocard. Among these are periodicals and pamphlets. It has been suggested by the Public Archives of Canada that microfiche could be used for the reproduction and dissemination of pamphlets as well as those publications of an archives itself that are pamphlet-length and that are out of print.

Initial production costs of preparing acetate jackets, aperture cards, microfiche, and microcards are so much higher than microfilm that on economic grounds alone it would be difficult to justify their use for archival materials.

Almost without exception archives have limited their use of microforms to roll microfilm for the reproduction of archival materials. It is recommended that this practice be continued.

In view of the foregoing this manual will be limited to a discussion of the use of roll microfilm.

7. <u>Administrative or facilitative uses.</u> Some series of records, for example, may be more easily used and the costs of reference service reduced by making microfilm copies of indexes, lists, or other finding aids to records when the original finding aids are required in the agency of origin, in another repository, or in two or more widely separated locations in a repository at the same time.

In the case, also, of self-indexing files of bulky dossiers that contain listings or descriptions of their contents on the covers of the dossiers, an index on film may be created by filming the dossier covers. Such an index may reduce reference to the dossier itself or it may expedite or reduce the cost of servicing the records. Another example, of an administrative or facilitative use is as an intermediate step in the preparation of microcards or microfiche or in the preparation of full size or slightly reduced copies of originals by the xerox copy flow process.

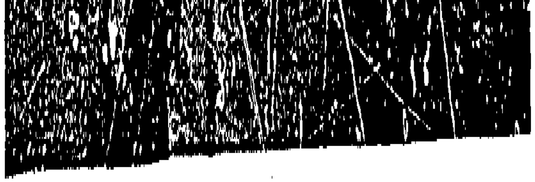
Microfilm may also be a part of the actual recording process.

III. DISADVANTAGES OF MICROFILM

Despite its significant uses microfilming should not be looked upon as a panacea for solving all problems for it has disadvantages as well as advantages. Some of the disadvantages of microfilming records are:

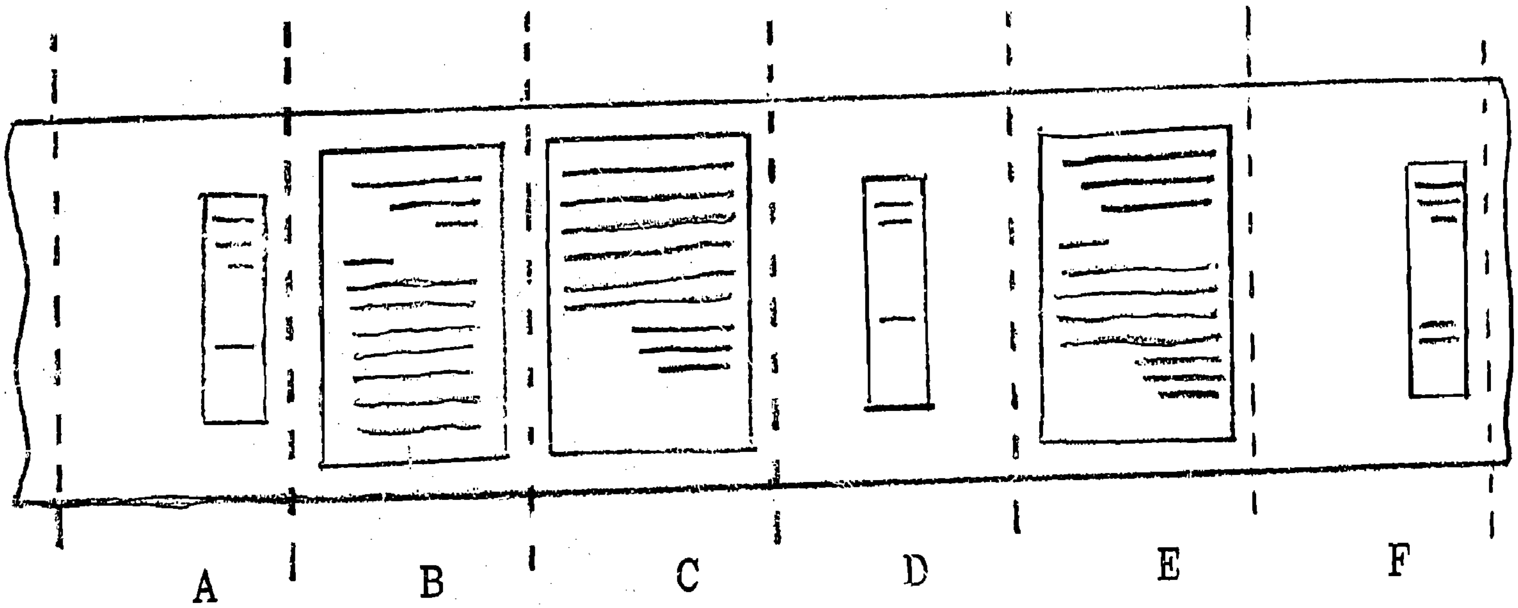
(1) the cost of maintaining and servicing records in microfilm form may be more expensive than storing and servicing the originals;
(2) the microfilm copy may be inconvenient to use;
(3) it is not always possible to obtain a perfectly legible or usable microfilm copy (a) when a document contains very fine writing, (b) when a document is exceptionally large (c) when there is extremely poor light-reflecting contrast between the reading matter and the paper, as in the case of some yellowish or faded inks or discolored, darkened, or colored paper, and (d) when the interpretation of a document depends upon color as a distinguishing element;
(4) the intrinsic value of a document is lost in the film copy;
(5) it is not possible to compare two separate images on the same roll;
(6) once records have been filmed it is not always feasible to incorporate additional material in the film;
(7) there is a danger that disposal microfilming will be used as a substitute for a thorough appraisal of records; and
(8) unless quality standards are carefully followed the microfilm produced may not serve its purpose.

ERIC



17. A microfilm enlarger

18. A reader-printer



A. When starting to film a folded document, film it unfolded endorsement side up on the right hand side of the camera's field

B&C. Unfold the document and film the pages in sequence as shown

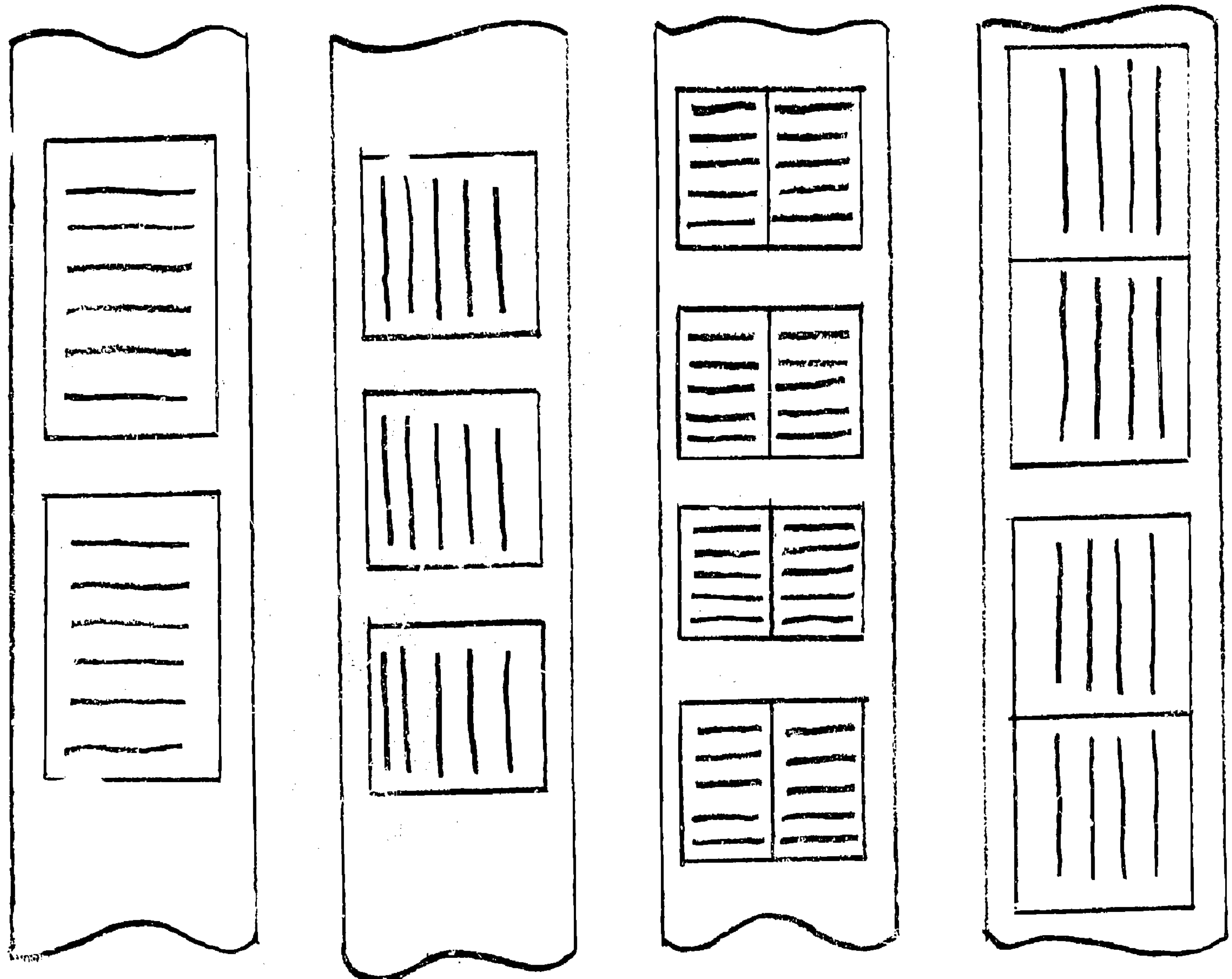
D. If the document contains one or more enclosures, center each folded enclosure in the middle of the field

E. Unfold the enclosure and film the pages in sequence

F. Film the endorsement on the right hand side of the field

19. A method for filming folded documents

ERIC Full Text Provided by ERIC



Placement IA   Placement IB   Placement IIA   Placement IIB

3. Standard Image Placements